
SUMMATION NOTATION

Full Solutions with Working & Reasoning
Problems 1 - 12

Sigma notation · Telescoping sums · Index shifting

01

WRITE USING SUMMATION SYMBOLS

Part (a) $a_3 + a_4 + a_5 + \dots + a_{n-1}$

The index starts at 3 and ends at $n-1$, stepping by 1. One index k running from 3 to $n-1$ captures every term.

$$\sum (k = 3 \text{ to } n-1) a_k$$

Part (b) $a_2 + a_4 + a_6 + \dots + a_{2n}$

All subscripts are even: 2, 4, 6, ..., $2n$. Writing the subscript as $2k$ with k from 1 to n captures each even-indexed term exactly.

$$\sum (k = 1 \text{ to } n) a_{2k}$$

Part (c) $a_3 + a_5 + a_7 + \dots$ (infinite)

Subscripts are odd starting at 3: the k -th odd number ≥ 3 is $2k+1$ for $k = 1, 2, 3, \dots$. The series does not terminate.

$$\sum (k = 1 \text{ to } \infty) a_{2k+1}$$

Part (d) $c_9 + c_8 + c_7 + \dots + c_2 + c_1$

The subscripts descend from 9 to 1. Addition is commutative, so we sum c_k for $k = 1$ to 9 (same terms, ascending order).

$$\sum (k = 1 \text{ to } 9) c_k$$

Part (e) $1 + 1/2 + 1/4 + 1/8 + \dots$ (infinite)

Each term equals $1/2^k$ for $k = 0, 1, 2, \dots$ — a geometric series with ratio $1/2$. The series is infinite.

$$\sum (k = 0 \text{ to } \infty) 1 / 2^k$$

Part (f) $n + (n-1) + (n-2) + \dots + 2 + 1$

The terms are the positive integers 1 through n in reverse order. Because addition is commutative we write k ascending from 1 to n .

$$\sum (k = 1 \text{ to } n) k$$

Part (g) $-1 + 2 - 3 + 4 - 5 + \dots$ (infinite)

Each term k carries the sign factor $(-1)^k$. At $k=1$ that gives $-1 \times 1 = -1$; at $k=2$ it gives $+1 \times 2 = +2$; and so on. The series is infinite.

$$\sum (k = 1 \text{ to } \infty) (-1)^k \cdot k$$

Part (h) $a_i + a_{i+2} + a_{i+4} + \dots + a_{i+2n}$

Subscripts form the sequence $i, i+2, i+4, \dots, i+2n$ — i.e. $i+2k$ for $k = 0, 1, \dots, n$. There are $n+1$ terms in total.

$$\sum (k = 0 \text{ to } n) a_{i+2k}$$

Part (i) $1/n + 2/(n+1) + 3/(n+2) + \dots + (n+1)/(2n)$

Numerators run $1, 2, \dots, n+1$ (i.e. k) while denominators run $n, n+1, \dots, 2n$ (i.e. $n+k-1$). The k -th term is $k/(n+k-1)$, $k = 1$ to $n+1$.

$$\sum (k = 1 \text{ to } n+1) k / (n + k - 1)$$

02

PARTIAL SUM USING THE TRIANGULAR-NUMBER FORMULA

Given $\sum(i=1 \text{ to } n) i = n(n+1)/2$, evaluate $\sum(i=34 \text{ to } n) i$

Reasoning

The formula $\sum(i=1 \text{ to } n) i = n(n+1)/2$ always starts at $i = 1$. Our sum starts at $i = 34$, so we cannot apply it directly. The trick is to subtract the "missing" part at the front:

$$\sum(i=34 \text{ to } n) i = \sum(i=1 \text{ to } n) i - \sum(i=1 \text{ to } 33) i$$

Both pieces now start at 1, so the formula applies to each:

$$= n(n+1)/2 - 33 \times 34/2$$

$$= n(n+1)/2 - 1122/2$$

$$\sum(i=34 \text{ to } n) i = n(n+1)/2 - 561$$

03

EVALUATE A DEFINITE SUM

Evaluate $\sum(j=40 \text{ to } 100) j$

Reasoning

We split off the terms from 1 to 39 so that each piece starts at $j = 1$, then apply the triangular-number formula and subtract:

$$\sum(j=40 \text{ to } 100) j = \sum(j=1 \text{ to } 100) j - \sum(j=1 \text{ to } 39) j$$

$$= (100 \times 101)/2 - (39 \times 40)/2$$

$$= 5050 - 780$$

$$\sum(j=40 \text{ to } 100) j = 4\,270$$

04

SUM OF A CONSTANT

Evaluate $\sum(k=10 \text{ to } 1000) 3$

Reasoning

When every term equals the same constant c , the total is simply $c \times$ (number of terms). The number of integers from $k = 10$ to $k = 1000$ inclusive is $1000 - 10 + 1 = 991$.

$$\sum_{k=10 \text{ to } 1000} 3 = 3 \times 991$$

$$= \mathbf{2\,973}$$

05

EXPAND AN ALTERNATING SUM

Expand $\sum(i=0 \text{ to } n) (-1)^i / (i+1)$

Reasoning

Substitute $i = 0, 1, 2, 3, \dots$ in turn. The factor $(-1)^i$ alternates: positive when i is even, negative when i is odd.

$$i = 0: (-1)^0 / (0+1) = +1/1 = 1$$

$$i = 1: (-1)^1 / (1+1) = -1/2$$

$$i = 2: (-1)^2 / (2+1) = +1/3$$

$$i = 3: (-1)^3 / (3+1) = -1/4$$

$\vdots$ and so on up to $i = n$

$$1 - 1/2 + 1/3 - 1/4 + \dots + (-1)^n/(n+1)$$

This is the partial sum of the alternating harmonic series.

06

TELESCOPING SUM

Expand and simplify $\sum(i=1 \text{ to } 5) (a_i - a_{i-1})$

Reasoning

Write out all five terms explicitly:

$$i=1: (a_1 - a_0)$$

$$i=2: (a_2 - a_1)$$

$$i=3: (a_3 - a_2)$$

$$i=4: (a_4 - a_3)$$

$$i=5: (a_5 - a_4)$$

When we add these, almost every term appears once positive and once negative — they cancel ("telescope"). Only the very first subtracted term and the very last added term survive:

$$= a_5 - a_0$$

$$\sum(i=1 \text{ to } 5) (a_i - a_{i-1}) = a_5 - a_0$$

07

SEPARATING THE LAST TERM

Rewrite $\sum(i=1 \text{ to } n+1) 1/i^2$ by separating off the last term

Reasoning

The sum runs $i = 1$ to $i = n+1$. We isolate the final term ($i = n+1$) so the sigma only runs to n :

$$\sum(i=1 \text{ to } n+1) 1/i^2 = \sum(i=1 \text{ to } n) 1/i^2 + [\text{term at } i = n+1]$$

$$= \sum(i=1 \text{ to } n) 1/i^2 + 1/(n+1)^2$$

08

MERGING A STRAY TERM INTO A SUMMATION

Rewrite $\sum(k=0 \text{ to } n) 2^k + 2^{n+1}$ as a single summation

Reasoning

The extra term 2^{n+1} is exactly the $k = n+1$ term of the geometric sequence 2^k — it is the very next term after where the sum ends. Extending the upper limit from n to $n+1$ absorbs it automatically:

$$\sum(k=0 \text{ to } n) 2^k + 2^{n+1}$$

$$= \sum(k=0 \text{ to } n) 2^k + \sum(k=n+1 \text{ to } n+1) 2^k$$

$$= \sum(k=0 \text{ to } n+1) 2^k$$

09

CHANGE OF SUMMATION VARIABLE

Technique: choose $j = f(i)$, recompute the lower and upper bounds under the new variable, then rewrite the summand entirely in terms of j .

Part (a) $\sum(i=1 \text{ to } n-1) i / (n-i)^2$, substitution $j = i-1$ (so $i = j+1$)

Step 1 — new lower bound: $i = 1 \rightarrow j = 1-1 = 0$

Step 2 — new upper bound: $i = n-1 \rightarrow j = (n-1)-1 = n-2$

Step 3 — rewrite the summand (replace every i with $j+1$):

numerator: $i = j+1$

denominator: $(n-i)^2 = (n-(j+1))^2 = (n-j-1)^2$

$$\sum(j=0 \text{ to } n-2) (j+1) / (n-j-1)^2$$

Part (b) $\sum(i=3 \text{ to } n) i/(i+n-1)$, substitution $j = i-1$ (so $i = j+1$)

Step 1 — new lower bound: $i = 3 \rightarrow j = 3-1 = 2$

Step 2 — new upper bound: $i = n \rightarrow j = n-1$

Step 3 — rewrite the summand (replace every i with $j+1$):

numerator: $i = j+1$

denominator: $i+n-1 = (j+1)+n-1 = j+n$

$$\sum(j=2 \text{ to } n-1) (j+1) / (j+n)$$

10

COMPUTING SUMS FROM A GIVEN SEQUENCE

Given: $a_1 = -2$, $a_2 = -1$, $a_3 = 0$, $a_4 = 1$, $a_5 = 2$

Part (a) $\sum_{i=1}^5 a_i$

Substitute each sequence value directly:

$$\begin{aligned} & a_1 + a_2 + a_3 + a_4 + a_5 \\ & = (-2) + (-1) + 0 + 1 + 2 \end{aligned}$$

$$= 0$$

The sequence is symmetric about 0 so positive and negative terms cancel exactly.

Part (b) $\sum_{i=0}^2 a_{2i+1}$

The subscript is $2i+1$. Substitute $i = 0, 1, 2$:

$$\begin{aligned} i=0: & a_1 = -2 \\ i=1: & a_3 = 0 \\ i=2: & a_5 = 2 \\ \text{Sum} & = (-2) + 0 + 2 \end{aligned}$$

$$= 0$$

11

SEPARATING THE LAST THREE TERMS

Rewrite $\sum_{i=1}^{n+1} 1/i^2$ by separating off the last three terms

Reasoning

The last three terms correspond to $i = n-1$, n , and $n+1$. We peel all three off, leaving the sigma to run only to $n-2$:

$$\begin{aligned} & \sum_{i=1}^{n+1} 1/i^2 \\ & = \sum_{i=1}^{n-2} 1/i^2 + (i=n-1 \text{ term}) + (i=n \text{ term}) + (i=n+1 \text{ term}) \end{aligned}$$

$$= \sum_{i=1}^{n-2} 1/i^2 + 1/(n-1)^2 + 1/n^2 + 1/(n+1)^2$$

12

SEPARATING FIRST AND LAST TERMS

Rewrite $\sum_{k=0}^n 2^k$ by separating off the first and last terms**Reasoning**

The first term is $k = 0$, giving $2^0 = 1$. The last term is $k = n$, giving 2^n . After removing both, the remaining sigma runs from $k = 1$ to $k = n-1$:

$$\sum_{k=0}^n 2^k = 2^0 + \sum_{k=1}^{n-1} 2^k + 2^n$$

$$= 1 + \sum_{k=1}^{n-1} 2^k + 2^n$$

END OF SOLUTIONS

12 problems · Summation notation · Full working shown